# Phil Hayward

Doncaster | 07883 892589 | Philip.Hayward@outlook.com | https://linkedin.com/in/pahayward

## Professional Summary

I'm an experienced senior/lead software engineer with thirty years of building secure, high integrity applications in multiple domains, using a range of technologies including C++17, C# and Python, on desktops, AWS, Azure, and docker containers. Proven ability to guide technical teams while maintaining a hands-on coding focus.

## Technical Skills

**AI:** Copilot/OpenCode & Cursor with cloud/local based LLM providers
**Languages & Frameworks:** C#, C++17, Python, JavaScript/TypeScript, React, WPF, ASP.NET Core
**Design Patterns:** MVC, MVVM, Observer, Command, Factory, Singleton, Adaptor, Decorator, Events
**Backend & Data:** .NET 9/Core, Entity Framework, REST/gRPC, SQL Server, PostgreSQL, Azure SQL
**Infrastructure & DevOps:** Linux, Docker, Azure (Linux VMs, App Services), AWS, Git, Jira, CI/CD
**Networking & Diagnostics:** IP networking, Wireshark, TCP/IP analysis, MPEG-2 streaming
**Security & Compliance:** Zero Trust, RBAC, TLS hardening, OWASP, GDPR-compliant data handling
**Practices:** TDD (xUnit), Microservices, MVC/MVVM, Scrum (remote teams), Observability Design

## Professional Experience

**Full-Stack Developer** | Brand 'n Deliver – Sheffield (Hybrid) | *Dec 2024 – Present*

- Integrated multiple third party B2B supplier APIs, extending products offering
- Integration of LLM models for copy/market material generation with OpenRouter
- Re-organized Azure infrastructure and migrated several components to docker with service bus/Signal R (event driven architecture which provides a more responsive user experience)
- Moved several services from manual deployment to bicep/PowerShell deployment
- Implemented observability using Application insights/Open Telemetry
- Added xUnit TDD, CoPilot & OpenCode Ai assistance to workflows.

**Lead Software Engineer** | Lockwood Publishing – Nottingham (Remote) | *Nov 2021 – Oct 2024*

- Lockwood publishing (free to play Android/IOS Game) operates on AWS, backed by Golang/Lua services with Unity game clients over TLS based encryption, worldwide with up to a million monthly active users.
- Led 4-person cross disciplinary team for implementation of GDPR/Consents management system implemented in Unity, as well as carrying out security hardening on AWS/Android client infrastructure with one junior report.
- Implemented performance enhancements for TCP/IP / REST stack & cache strategies

**Senior Software Engineer |** Certara (Simcyp) – Sheffield | *Nov 2008 – Nov 2021*

- Built out the QA function (5-person test team) and worked with science team to build out several subsystems of a £300k/license pharmaceutical trials modelling application which enables the world top pharma to save millions on clinical trials. Also led development of anonymized clinical trials data collaboration tools used by a pharmaceutical consortium project.
- Tech stack: C++ (ODE simulations), C#/WPF (UI) and COM/VBA (Excel automation) as well as multiple output targets for data warehousing (ODB drivers), Azure/Web app with SQL Server

**Senior Software Engineer (Security & Surveillance) |** Synectics Systems| *2001 – 2008*

- Enabled key technology business drivers for several high-profile command and control projects in digital CCTV which includes Manchester City Centre, London's ring of Steel, Rolls Royce.
- Active supporting bid process & technical prototypes development.

## Publication

Hayward P, et al. "Biopharmaceutics data management system for anonymised data sharing and curation: First application with OrBiTo IMI project." *Comput Methods Programs Biomed.* 2017;140:1–10. PMID: 28254086

## Education

**BSc (Hons) Software Engineering (2:1)**
Sheffield Hallam University 2.1 with specialization C/Unix/Image Processing | 1996

## Continuous Learning & Certifications

• Docker, GitOps & AI/ML (self-study), AWS, Azure & Data Science

• Linux System Administration (self-study), Agentic AI systems & ML